Berkelium

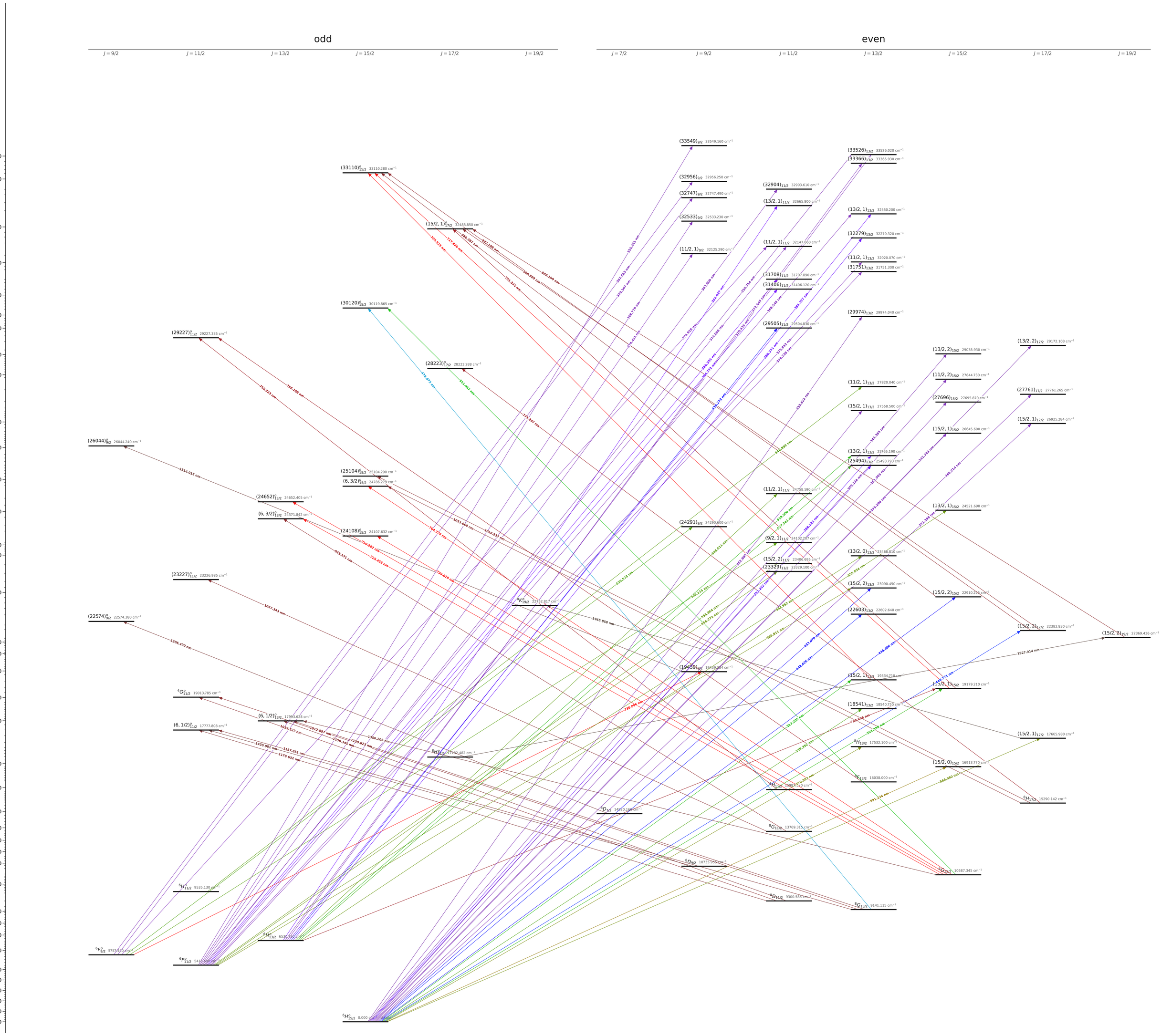
Bk (Bk I). Classified transitions below 2 µm (levels and lines from the literature): 80 levels, 86 lines.

Bk^{+} ionisation limit 49989.00 cm^{-1} = 6.19785 eV (200.044 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



^{249}Bk hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^{6}F^{o}_{11/2}$	590.600	1061.300	F=2: -12997.16 F=3: -11399.02 F=4: -9229.59 F=5: -6455.77 F=6: -3036.22 F=7: +1078.69 F=8: +5946.83 F=9: +11634.38
$^{8}G_{13/2}$	524.600	4586.800	F=3: -12006.60 F=4: -10714.67 F=5: -8910.74 F=6: -6468.80 F=7: -3237.64 F=8: +959.16 F=9: +6323.22 F=10: +13081.35
$^{8}D_{11/2}$	575.600	3978.200	F=2: -11449.37 F=3: -10373.55 F=4: -8794.46 F=5: -6588.10 F=6: -3599.49 F=7: +357.38 F=8: +5499.50 F=9: +12074.85
$^{8}G_{15/2}$	512.600	7635.700	F=4: -12468.27 F=5: -11307.75 F=6: -9572.29 F=7: -7074.91 F=8: -3597.43 F=9: +1109.47 F=10: +7326.28 F=11: +15364.67
$^{8}D_{9/2}$	638.600	8993.800	F=1: -8170.89 F=2: -8178.52 F=3: -7922.29 F=4: -7080.99 F=5: -5226.36 F=6: -1823.04 F=7: +3771.39 F=8: +12306.40
$^{8}G_{11/2}$	407.700	-104.900	F=2: -9318.57 F=3: -8078.30 F=4: -6428.43 F=5: -4372.22 F=6: -1913.76 F=7: +942.05 F=8: +4189.48 F=9: +7822.00
$^{8}H_{17/2}$	560.600	12630.300	F=5: -14228.63 F=6: -13253.11 F=7: -11511.35 F=8: -8724.75 F=9: -4574.88 F=10: +1296.46 F=11: +9287.29 F=12: +19835.42
$^{8}H_{11/2}$	506.600	3564.500	F=2: -10050.74 F=3: -9114.23 F=4: -7735.92 F=5: -5804.72 F=6: -3181.74 F=7: +299.65 F=8: +4833.89 F=9: +10643.17
$^{8}F_{13/2}$	422.700	2443.300	F=3: -10156.14 F=4: -8894.93 F=5: -7217.74 F=6: -5057.43 F=7: -2333.46 F=8: +1048.14 F=9: +5194.77 F=10: +10227.25
(15/2, 0)$_{15/2}$	1710.300	6955.200	F=4: -48347.75 F=5: -41073.73 F=6: -32032.64 F=7: -21054.14 F=8: -7939.51 F=9: +7538.35 F=10: +25634.96 F=11: +46634.18
$^{8}H^{o}_{17/2}$	2134.500	3798.400	F=5: -69645.48 F=6: -57556.66 F=7: -43271.50 F=8: -26706.21 F=9: -7765.03 F=10: +13659.77 F=11: +37687.88 F=12: +64450.97
$^{8}H_{13/2}$	98.900	6595.400	F=3: -59.43 F=4: -823.46 F=5: -1506.71 F=6: -1927.99 F=7: -1869.86 F=8: -1078.66 F=9: +735.52 F=10: +3898.82
(15/2, 1)$_{17/2}$	2593.200	797.400	F=5: -85945.40 F=6: -70536.96 F=7: -52522.35 F=8: -31883.97 F=9: -8601.70 F=10: +17347.05 F=11: +45987.42 F=12: +77347.05
(6, 1/2)$^{e}_{11/2}$	140.900	7305.900	F=2: -183.49 F=3: -956.30 F=4: -1721.05 F=5: -2250.01 F=6: -2258.55 F=7: -1405.08 F=8: +708.89 F=9: +4538.80
(6, 1/2)$^{e}_{13/2}$	1025.300	4688.800	F=3: -25110.74 F=4: -21833.94 F=5: -17544.73 F=6: -12114.28 F=7: -5388.03 F=8: +2814.37 F=9: +12699.02 F=10: +24497.78
(18541)$_{13/2}$	-215.900	4952.600	F=3: +7572.22 F=4: +5837.83 F=5: +3873.94 F=6: +1816.60 F=7: -170.91 F=8: -1898.11 F=9: -3147.30 F=10: -3673.57
$^{4}G^{o}_{11/2}$	620.600	3924.300	F=2: -12495.42 F=3: -11275.77 F=4: -9506.88 F=5: -7066.43 F=6: -3801.51 F=7: +471.34 F=8: +5966.17 F=9: +12927.62
(15/2, 1)$_{15/2}$	-155.900	9932.100	F=4: +8256.15 F=5: +5652.38 F=6: +2973.80 F=7: +463.63 F=8: -1594.36 F=9: -2875.84 F=10: -3015.97 F=11: -1609.35
(15/2, 1)$_{13/2}$	-311.800	4505.900	F=3: +9917.79 F=4: +7878.34 F=5: +5514.72 F=6: +2950.70 F=7: +334.84 F=8: -2159.56 F=9: -4334.44 F=10: -5966.98
(19439)$_{9/2}$	479.700	3522.600	F=1: -7619.70 F=2: -7163.53 F=3: -6374.43 F=4: -5126.60 F=5: -3252.30 F=6: -541.84 F=7: +3256.38 F=8: +8435.92
(15/2, 2)$_{19/2}$	2332.400	14569.900	F=6: -80795.16 F=7: -67280.10 F=8: -51103.99 F=9: -31974.71 F=10: -9563.61 F=11: +16494.47 F=12: +46601.20 F=13: +81194.77
(15/2, 2)$_{17/2}$	1987.600	6625.400	F=5: -63773.68 F=6: -53100.78 F=7: -40332.41 F=8: -25322.42 F=9: -7903.79 F=10: +12111.40 F=11: +34931.93 F=12: +60787.45
(22574)$^{e}_{9/2}$	326.800	1906.700	F=1: -5417.00 F=2: -5035.78 F=3: -4407.21 F=4: -3463.19 F=5: -2112.93 F=6: -242.92 F=7: +2283.01 F=8: +5623.77
(22603)$_{13/2}$	998.300	7273.000	F=3: -23408.07 F=4: -20693.64 F=5: -17000.89 F=6: -12130.01 F=7: -5841.24 F=8: +2145.16 F=9: +12148.88 F=10: +24529.58
$^{8}K^{o}_{19/2}$	1103.800	9812.200	F=6: -37250.88 F=7: -31417.86 F=8: -24259.72 F=9: -15579.71 F=10: -5156.51 F=11: +7255.80 F=12: +21927.74 F=13: +39154.40
(15/2, 2)$_{15/2}$	1166.200	4658.800	F=4: -32997.32 F=5: -28022.01 F=6: -21842.48 F=7: -14344.62 F=8: -5395.33 F=9: +5157.51 F=10: +17485.06 F=11: +31777.45
(23227)$^{e}_{11/2}$	1004.000	1199.200	F=2: -22344.97 F=3: -19529.20 F=4: -15731.24 F=5: -10913.70 F=6: -5029.86 F=7: +1976.33 F=8: +10170.30 F=9: +19626.80
(23329)$_{11/2}$	260.800	4556.800	F=2: -4048.34 F=3: -4011.60 F=4: -3796.91 F=5: -3262.24 F=6: -2230.05 F=7: -487.30 F=8: +2214.56 F=9: +6159.60
(15/2, 2)$_{11/2}$	656.500	4182.100	F=2: -13205.51 F=3: -11920.35 F=4: -10054.73 F=5: -7478.30 F=6: -4028.12 F=7: +491.34 F=8: +6308.20 F=9: +13683.15
(13/2, 0)$_{13/2}$	1148.200	3978.200	F=3: -28610.17 F=4: -24716.84 F=5: -19686.23 F=6: -13409.06 F=7: -5754.18 F=8: +3431.42 F=9: +14322.60 F=10: +27116.10
(24108)$^{e}_{15/2}$	765.700	3732.400	F=4: -21419.92 F=5: -18276.96 F=6: -14337.83 F=7: -9511.13 F=8: -3690.22 F=9: +3246.78 F=10: +11436.99 F=11: +21032.72
(6, 3/2)$^{e}_{13/2}$	293.800	2961.900	F=3: -6573.06 F=4: -5918.63 F=5: -4978.54 F=6: -3671.42 F=7: -1899.62 F=8: +450.78 F=9: +3509.97 F=10: +7424.43
(13/2, 1)$_{15/2}$	2824.000	599.600	F=4: -83795.57 F=5: -69785.71 F=6: -52946.94 F=7: -33264.60 F=8: -10721.55 F=9: +14701.80 F=10: +43027.45 F=11: +74279.90
(24652)$^{e}_{13/2}$	722.500	4916.600	F=3: -17074.62 F=4: -15049.08 F=5: -12314.55 F=6: -8735.95 F=7: -4151.20 F=8: +1628.80 F=9: +8820.17 F=10: +17666.03
(11/2, 1)$_{11/2}$	-566.600	3330.700	F=2: +14267.85 F=3: +12023.02 F=4: +9151.04 F=5: +5755.72 F=6: +1966.82 F=7: -2059.94 F=8: -6142.88 F=9: -10074.38
(6, 3/2)$^{e}_{15/2}$	1151.200	9263.600	F=4: -30873.60 F=5: -26819.08 F=6: -21537.74 F=7: -14802.71 F=8: -6349.32 F=9: +4124.91 F=10: +16960.28 F=11: +32534.90
(25104)$^{e}_{15/2}$	1693.200	4415.900	F=4: -48764.05 F=5: -41109.13 F=6: -31724.97 F=7: -20503.41 F=8: -7318.29 F=9: +7974.58 F=10: +25537.42 F=11: +45550.47
(25494)$_{13/2}$	536.600	5444.200	F=3: -11991.83 F=4: -10802.65 F=5: -9091.83 F=6: -6709.80 F=7: -3477.08 F=8: +815.72 F=9: +6407.91 F=10: +13568.70
(13/2, 1)$_{13/2}$	631.100	8274.300	F=3: -13383.95 F=4: -12314.37 F=5: -10636.43 F=6: -8122.80 F=7: -4500.70 F=8: +548.10 F=9: +7387.31 F=10: +16426.10
(26044)$^{e}_{9/2}$	377.700	8694.000	F=1: -3285.97 F=2: -3772.57 F=3: -4243.73 F=4: -4388.93 F=5: -3794.18 F=6: -1941.97 F=7: +1788.67 F=8: +8122.27
(15/2, 1)$_{15/2}$	272.800	4047.200	F=4: -6641.46 F=5: -6020.83 F=6: -5094.35 F=7: -3762.92 F=8: -1910.91 F=9: +593.85 F=10: +3900.02 F=11: +8172.80
(15/2, 1)$_{17/2}$	-365.700	9815.200	F=5: +15587.63 F=6: +11537.61 F=7: +7281.70 F=8: +3036.41 F=9: -950.82 F=10: -4401.62 F=11: -7006.68 F=12: -8425.77
(15/2, 1)$_{13/2}$	809.400	6781.300	F=3: -18638.56 F=4: -16593.27 F=5: -13757.22 F=6: -9944.10 F=7: -4930.35 F=8: +1544.85 F=9: +9779.58 F=10: +20109.17
(27761)$_{17/2}$	2020.600	2590.200	F=5: -66280.29 F=6: -54646.43 F=7: -40949.80 F=8: -25133.26 F=9: -7131.52 F=10: +13128.90 F=11: +35729.61 F=12: +60760.40
(28223)$^{e}_{17/2}$	2662.200	638.600	F=5: -88295.11 F=6: -72442.65 F=7: -53917.60 F=8: -32705.86 F=9: -8791.34 F=10: +17844.08 F=11: +47220.51 F=12: +79360.10
(13/2, 2)$_{15/2}$	1924.700	6439.500	F=4: -54914.01 F=5: -46473.27 F=6: -36055.27 F=7: -23502.30 F=8: -8630.37 F=9: +8770.78 F=10: +28937.71 F=11: +52133.25
(13/2, 2)$_{17/2}$	2401.300	10723.600	F=5: -76097.85 F=6: -63717.62 F=7: -48761.50 F=8: -30992.92 F=9: -10141.56 F=10: +14096.72 F=11: +42059.86 F=12: +74119.57
(29227)$^{e}_{11/2}$	629.600	3477.600	F=2: -12884.94 F=3: -11565.20 F=4: -9679.09 F=5: -7118.22 F=6: -3747.09 F=7: +596.88 F=8: +6103.38 F=9: +12989.20
(15/2, 1)$^{e}_{17/2}$	3012.900	7638.700	F=5: -97511.00 F=6: -80877.89 F=7: -61107.51 F=8: -38031.36 F=9: -11456.87 F=10: +18832.60 F=11: +53077.78 F=12: +91543.45
(33110)$^{e}_{15/2}$	2305.100	7282.000	F=4: -65924.00 F=5: -55736.01 F=6: -43183.47 F=7: -28088.06 F=8: -10241.71 F=9: +10593.36 F=10: +34684.65 F=11: +62329.38

Measured A, B constants; sources in the data file.